

José María Barbeito

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EDUCATION

North Carolina State University, *Raleigh, NC*

- **B.S.E:** Material Science Engineering; **Minor:** Chemistry *Expected May 2025*
- **Current Cumulative GPA:** 3.7/4.0

Trinity School of Durham and Chapel Hill (TSDCH), *Durham, NC*

- High School Diploma *April 2020*
- **Cumulative GPA:** 3.7/4.0

INDEPENDENT PROJECT EXPERIENCE

Aquarium and Vivarium Construction, Fabrication Engineer, *Durham, NC*

March 2018 - Present

- Constructed a 55 gallon poison dart frog vivarium to house five distinct varieties of poison dart frogs, designed to be a self-sustaining closed cycle.
- Created a 28 gallon aquarium based around crystalline structures of natural gemstones such as amethyst geodes, quartz clusters, and copper sulfate crystals.
- Applied wet lab experience to create copper sulfate crystal structures at home.

BEE Educated, Founder, *Durham, NC*

August 2014 – Dec 2020

- Founder and lead instructor of BEE Educated, a local outreach program that teaches children ages 8-12 the importance of bees through an on-site program with active apiaries.
- Raised bees for eight years, doing inspections and honey harvesting, processing, and selling to local businesses.
- Helped local beekeepers with their own hives, teaching and assisting them in keeping healthy colonies, along with providing swarm capture services for locals.

WORK EXPERIENCE

Mesas, Entrepreneur, *Durham, NC*

February 2018 – Present

- Started a venture selling custom river tables and resin cutting boards out of oak, walnut, ash, and maple.
- Designed and built a 10' long outdoor dining table out of dyed red epoxy resin and live edge white oak, working alongside a local blacksmith to fabricate a steel trestle base.

Freelance Contractor, *Durham, NC*

August 2019 – Present

- Worked locally to design and execute a variety of projects, depending on client needs.
- Projects include barn doors, chicken coops, garden fences, bunk beds, benches, tables, electrical installations, bathroom tiling, porch renovations, and concrete repair.

LEADERSHIP AND COMMUNITY EXPERIENCE

TSDCH Chemistry Club, Leader and Co-founder, *Durham, NC*

August 2019 – April 2020

- Founded and led the Chemistry Club at TSDCH.
- Organized a program for first graders to learn about phase changes through super-saturated solutions and gas-producing reactions.

TSDCH Cross Country and Track, Captain, *Durham, NC*

August 2019 – April 2020

- Led the TSDCH cross country team as Captain during the school's first regional conference win.
- Won 3rd place in NC State Championship meet for 300m hurdles.

TECHNICAL SKILLS: Woodworking, Metalworking, Lathe, Crystalline Structure Fabrication, Drone/Digital Cinematography

SOFTWARE SKILLS: JavaScript, Adobe Lightroom, Adobe Photoshop, Video Editing and Production

LANGUAGES: Spanish (Native), German (Beginner)

INTERESTS: Computer Assembly, Backpacking, Traveling, Rock Climbing, Scuba Diving, Drums, Guitar, Crystal Growing